



HPC

Access/Connect

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MesoFC provides access to two HPC cluster:

1. **Lumiere** is the historical cluster which provides the default computations resources
2. **Helios** is the new cluster designed deep-learning/machine-learning applications and for some modern parallel applications.

The two clusters are completely isolated from each other.

WARNING

The access to helios cluster requires an account and an explicit request.

Please contact us for more information.

SSH access

WARNING

VS Code Remote-SSH and similar tools are not allowed on MesoFC clusters.

Login nodes are **not development machines**. They are shared HPC resources used for login, compilation, and job submission.

Remote IDE tools start background services, consume shared resources, and impact other users.

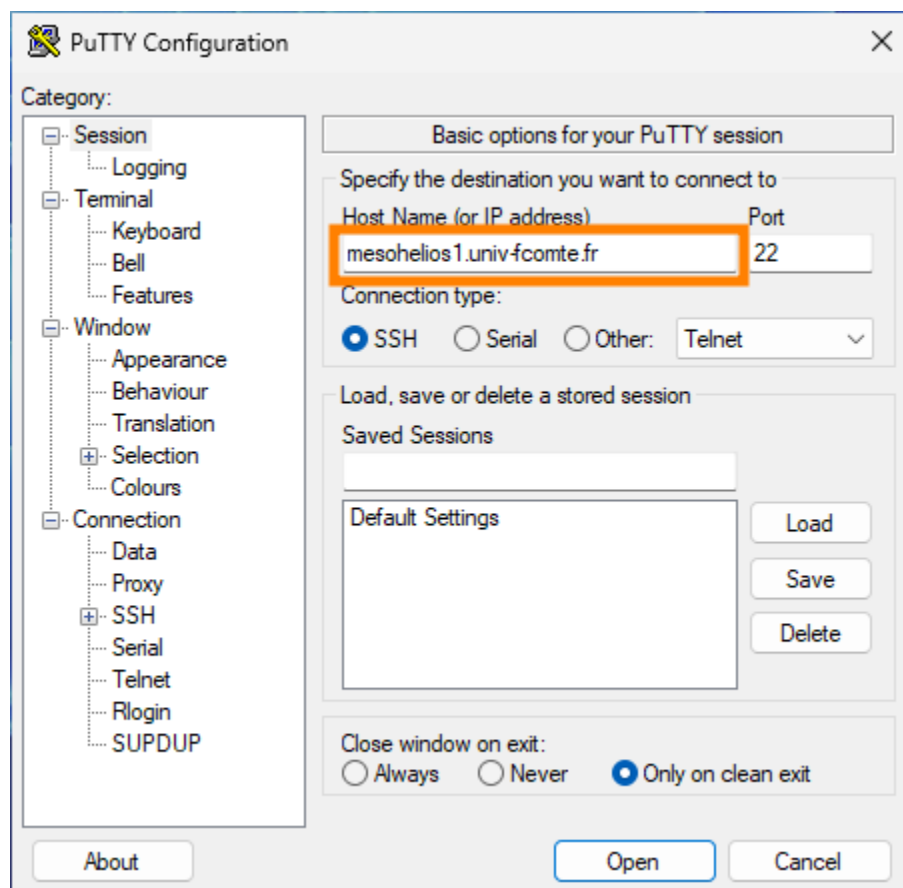
Such processes may be automatically killed.

Once your account is created and **authorized** to use Helios cluster, you can connect using SSH protocol to the frontal node `mesohelios1.univ-fcomte.fr`.

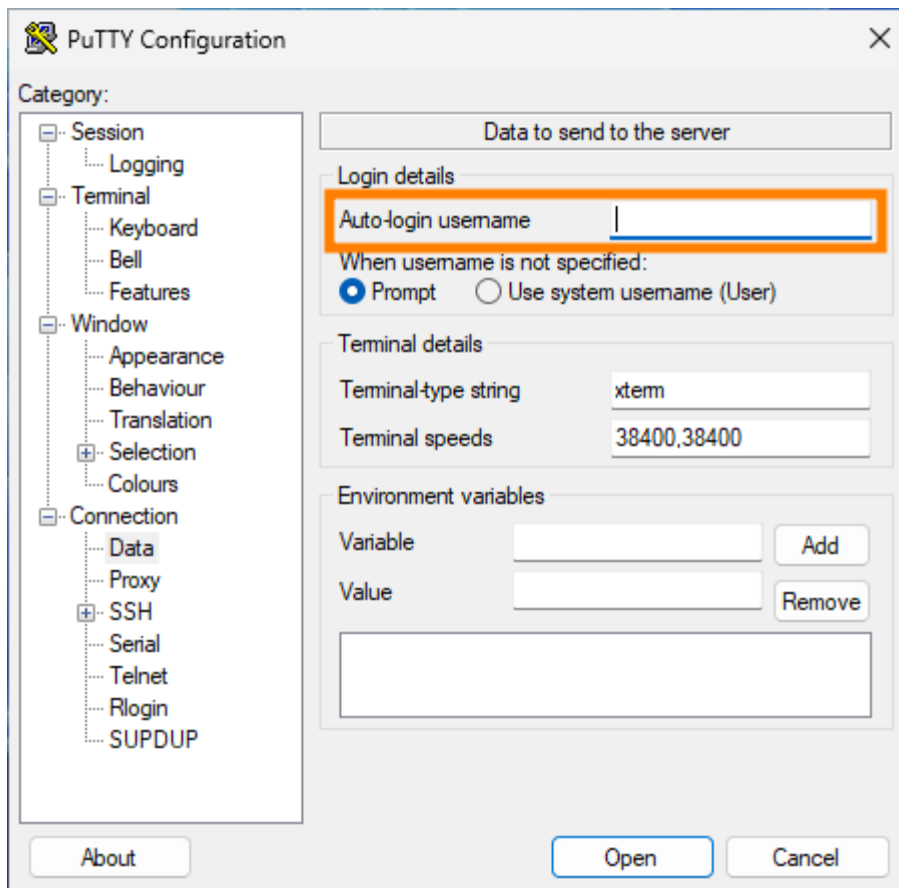
GNU Linux Windows Mac OS X

You will need an SSH client that supports the SSH-2 protocol: e.g. `Bitvise`, `OpenSSH`, `PuTTY`, or `SecureCRT`.

We will take PuTTY as an example:

**💡 AUTO-LOGIN USERNAME**

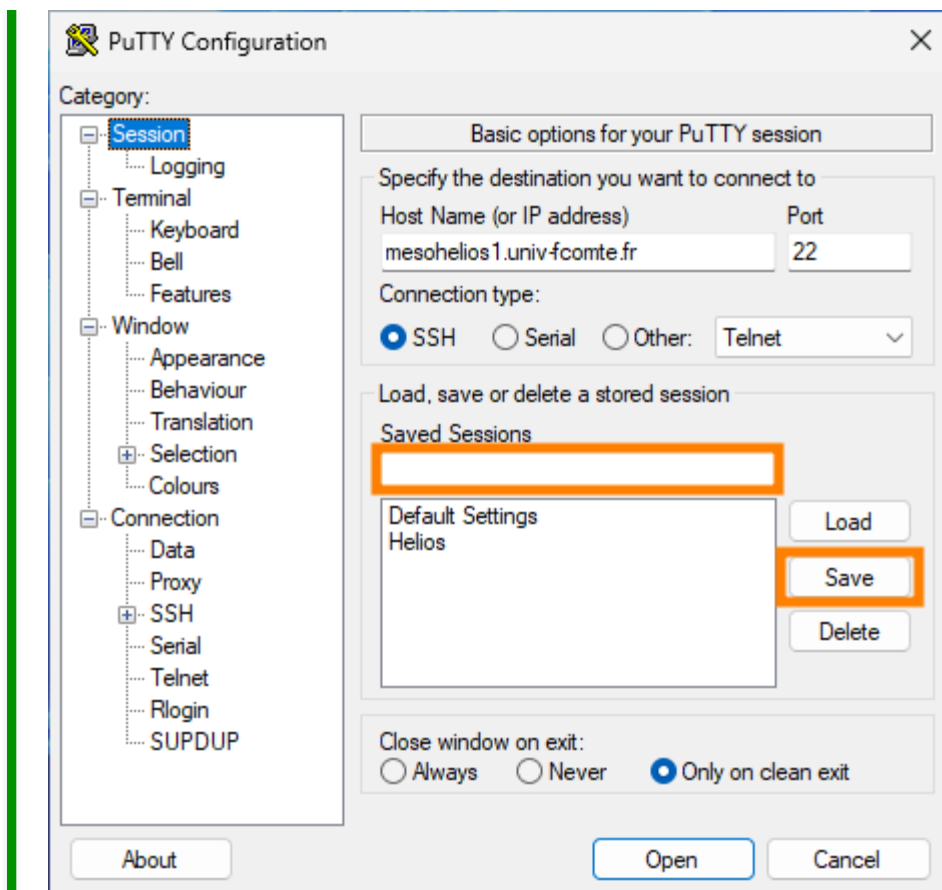
In the `Connection` → `Data` category, you can put your username in `Auto-login username` field so that PuTTY does not prompt you for it.



💡 SAVE SESSION

You can also save your current settings by simply entering the name you want in the field (framed in orange in screenshot) and clicking **Save**.

You can see that a session has already been saved in the screenshot : **Helios**.



Graphical access

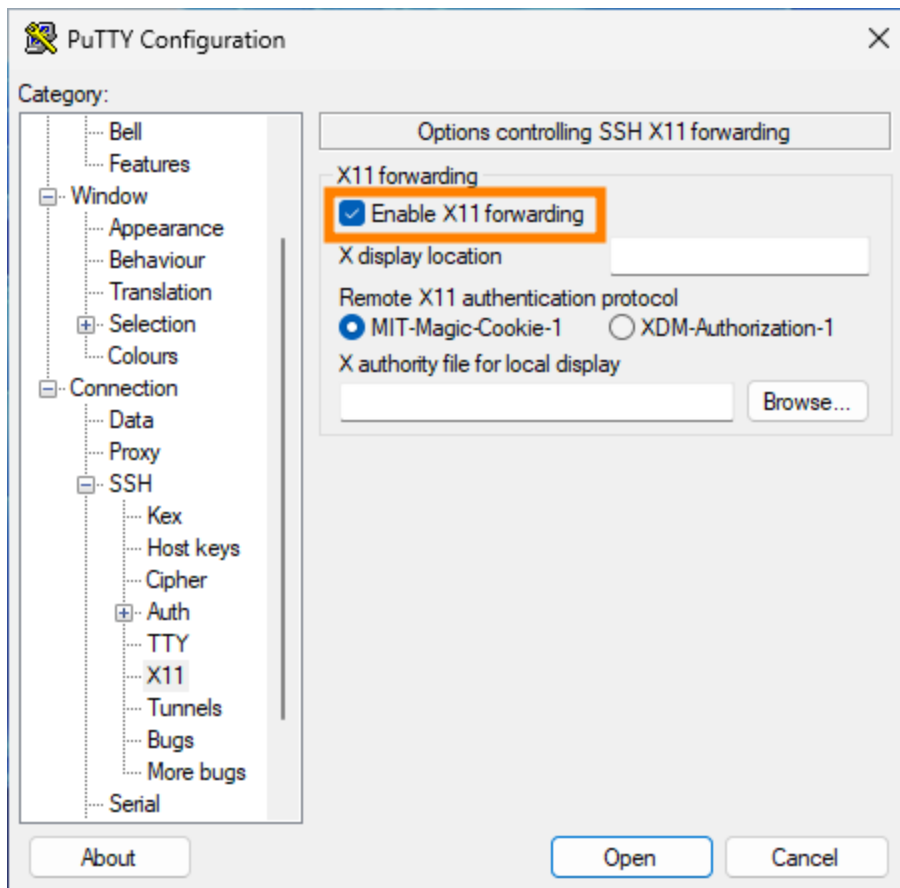
Using SSH

GNU Linux **Windows** Mac OS X

Windows does not provides an X server, so you have to download and install one: e.g. `Xming` or `VcXsrv`.

On your SSH client, you will also have to configure `X11 forwarding`.

For example for PuTTY, the `X11 forwarding` settings are inside `Connection` → `SSH` → `X11` category.



Using X2GO

An alternative is X2GO, you can download it [here](#) and use it as follows:

Paramétrage de la session - Helios

Session Connexion Entrées/Sorties Media Dossiers partagés

Nom de la session : Helios

 << changer d'icône

Chemin : /

Serveur

Hôte : mesohelios1.univ-fcomte.fr

Identifiant :

Port SSH : 22

Utiliser une clé RSA/DSA pour la connexion SSH :

☐ Essayer l'authentification automatique (via l'agent SSH ou la clé SSH par défaut)

☐ Authentification Kerberos 5 (GSSAPI)

☐ Délégation de l'identification GSSAPI au serveur

☐ Utiliser un serveur mandataire pour la connexion SSH

Type de session

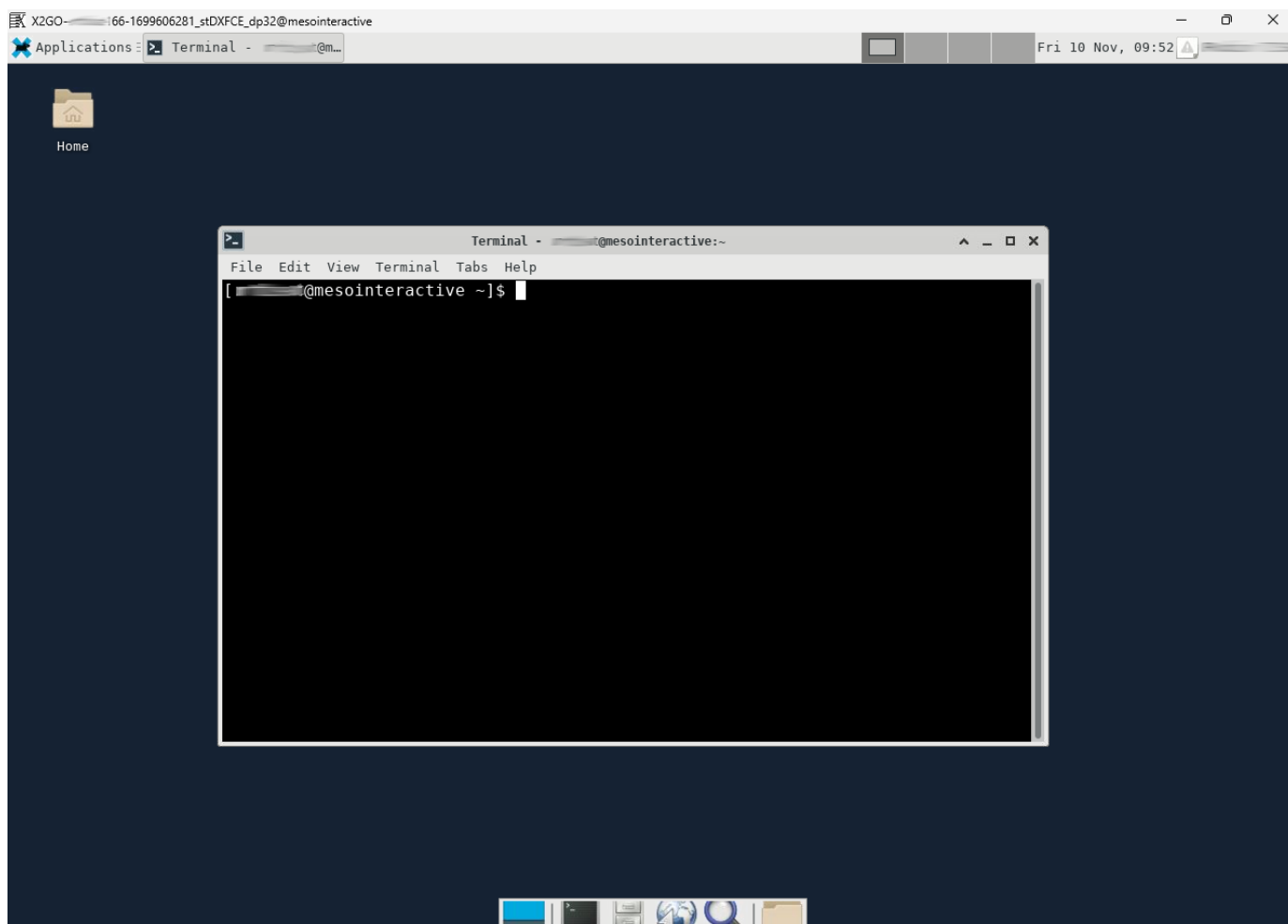
☐ Démarrer dans X2GoKDrive (expérimental)

XFCE

Commande :

Ok Cancel Défauts

Once connected, we are in XFCE desktop environment:



To log out from XFCE session, you have two choices:

- Click on the **Log Out** menu item on the **Applications** menu and confirm by clicking on the **Log Out** button.
- Click on the **Log Out...** menu item on the right top menu and confirm by clicking on the **Log Out** button.

If you want to keep your session alive, simply close your X2GO window, your session will be resumed the next time you log in with X2GO.

WARNING

The frontal node `mesohelios1.univ-fcomte.fr` is restarted every day at 04h00 am, so your remaining sessions will be killed.

VPN Access

Helios is only accessible from the university network. Access from outside is via VPN.

Please contact your IT manager in your laboratory or contact the university VPN team directly, [their website](#).